

# Passive GPON Monitoring for Network Load Forecasting Using BWMaP Information

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**Abstract** We present a vendor and element management system-agnostic solution for passive upstream bandwidth monitoring in GPON. We demonstrate its usage in a commercial platform for forecasting the network congestion state and providing services profiling, enabling SLA verification and identification of low-congestion windows. ©2026 The Author(s)

## Introduction

Gigabit Passive Optical Networks (GPON) [1] enable multiple users to share a common fiber in a point-to-multipoint topology through time division multiplexing/multiple access (TDM/TDMA) over a single downstream/upstream wavelength pair. In GPON, the downstream operates in continuous mode using a synchronous transmission structure with a fixed 125  $\mu$ s period, with the OLT broadcasting data streams to all ONUs. Each downstream frame carries not only user data but also control plane information, including the upstream Bandwidth Map (BWMaP) embedded in the GPON Transmission Convergence (GTC) header, which specifies the transmission schedule for all ONUs. Indeed, upstream communication occurs in burst mode, with each ONU transmitting only within its assigned time slots informed by the BWMaP. These slots are dynamically allocated by the OLT according to service requirements and network congestion and by means of vendor-specific dynamic bandwidth allocation (DBA) algorithms [2].

This upstream scheduling information is a remarkably rich signal: broadcast every frame without encryption, the BWMaP encodes the complete upstream allocation schedule for all ONUs in the network. Yet, field GPON deployments remain largely opaque to external observers as upstream utilization analysis typically requires proprietary OLT element management systems (EMS), limiting the ability of researchers, auditors and service providers to independently verify service-level agreement (SLA) compliance or detect equipment anomalies. Passive recovery of the BWMaP from the downstream signal circumvents this barrier entirely, enabling vendor-independent congestion monitoring without any sort of intervention at the network equipment side.

Prior work on GPON monitoring has explored passive optical-layer capture [3] and SDN control-plane interception [4], but neither recovers upstream scheduling information from the BWMaP.

Applying the traffic forecasting techniques studied in IP, mobile and optical networks [5, 6] to this passively recovered signal for vendor-agnostic congestion prediction in GPON remains unexplored.

In this work, we present a solution for passively capturing and decoding GPON downstream frames, extracting BWMaP allocations and applying forecasting techniques to identify non-congested upstream utilization windows. We demonstrate that this signal is reliable and privacy-preserving, enabling real-time congestion monitoring, DBA profiling and the prediction of low-utilization periods in live GPON deployments.

## Signal Acquisition and Demodulation

This work builds upon a fully operational GPON testbed deployed in our laboratory using commercial-grade equipment [7]. This infrastructure has been successfully used in prior studies for DBA analysis and reactive transmission container (T-CONT)[8, 9].

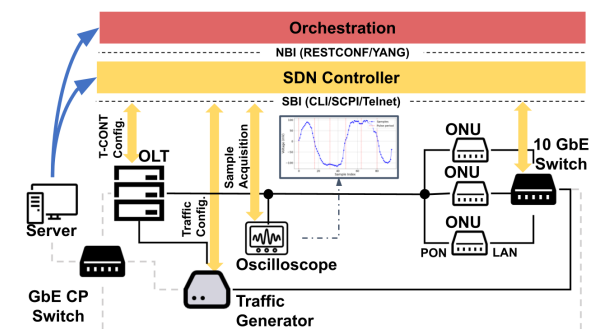


Fig. 1: Testbed diagram with the SDN layer and orchestration abstraction.

The testbed (Fig. 1) consists of one OLT, multiple ONUs, a 1:32 passive optical splitter, data switches, an oscilloscope, an Ethernet traffic generator, and a dedicated server, all managed through a RESTCONF/YANG SDN control layer.

A diplexer placed between the OLT and the passive splitter isolates the downstream wavelength, which is then converted into an electrical signal by a photodiode and fed into a high-speed oscillo-

scope. The oscilloscope is directly connected to the SDN controller and the raw samples are stored on a central server for offline processing. A fixed sampling window of 5 ms is used here with an acquisition sampling rate of 5 GS/s. The captured raw signals are processed using a custom Python script based on the GNURadio framework [10]. GNURadio was selected for its extensive library of ready-to-use signal processing blocks, which simplify the extraction of bit sequences from the captured signals.

Demodulation consists of four sequential steps: filtering, resampling, synchronization, and threshold-based decision making. The resulting bit stream is then frame-synchronized using the preamble, descrambled according to the standard, and decoded to recover the Physical Control Block downstream (PCBd) header. This allows us to extract the start and stop bytes assigned to each allocation ID in the network, corresponding to the services of each ONU, directly from raw samples via a purely passive monitoring approach.

### Results and Discussion

Before trying to do any sort of congestion prediction, we need to understand the circumstances under which the upstream allocation grants of each ONU that we measure inside the GTC headers translate to actual upstream traffic generated by our Ethernet generator. To do so, we analyse the behaviour of the network allocation for different traffic and T-CONT configurations through an experimental approach.

Three ONUs were configured with T-CONT 3, each with a PIR of 1.24 Gbps and a CIR of 50 Mbps, meaning each ONU can achieve up to 1.24 Gbps when resources are available, while the network guarantees a minimum of 50 Mbps when needed. We defined three traffic scenarios: T1, T2, and T3. For each scenario, we report the traffic injected at the ONU (Tx), the traffic received at the OLT (Rx), and the transmission window granted by the OLT (Tx Grant).

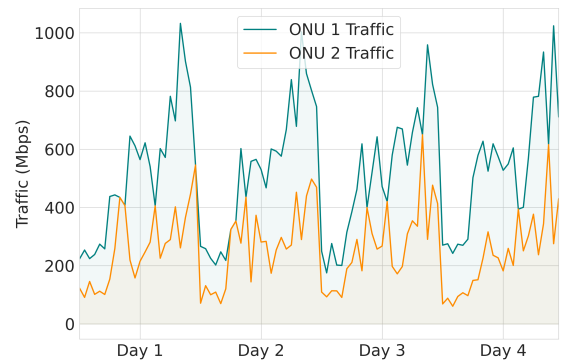
**Tab. 1:** Results of the experiment (values in Mbps)

| Scenario | ONU | Tx    | Rx    | Tx Grant |
|----------|-----|-------|-------|----------|
| T1       | 1   | 486.3 | 404.2 | 411.8    |
|          | 2   | 486.3 | 404.2 | 411.8    |
|          | 3   | 486.3 | 404.2 | 411.8    |
| T2       | 1   | 97.2  | 97.2  | 99.0     |
|          | 2   | 972.6 | 624.5 | 640.6    |
|          | 3   | 486.3 | 486.3 | 495.7    |
| T3       | 1   | 97.2  | 97.2  | 296.5    |
|          | 2   | 48.6  | 48.6  | 246.2    |
|          | 3   | 486.3 | 486.3 | 692.5    |

The results, shown in Table 1, demonstrate fair bandwidth sharing among the three ONUs when their demands are similar. In scenario T1, all ONUs inject traffic well above the available

grants and receive nearly identical transmission windows ( $\sim 411.8$  Mbps), with the received traffic (Rx) closely matching the granted bandwidth. In T2, ONU2 significantly increases its transmitted load. The DBA responds by allocating it a larger grant (640.6 Mbps); however, due to limited resources, this grant does not fully satisfy ONU2's demand, and the DBA instead prioritizes the lower demands of ONU1 and ONU3. In T3, the aggregate demand falls below the total upstream capacity, and ONU3 receives the highest Tx Grant (692.5 Mbps) thanks to its higher transmitted load, illustrating the dynamic and demand-aware behavior of the DBA under T-CONT 3. Nevertheless, because the network is not congested and the DBA grants bandwidth even when users do not need it, the traffic estimations are less precise.

Overall, the results indicate that the Tx Grant accurately estimates the received data rate mainly when the network operates near capacity. When the aggregate demand is significantly below the available capacity, the investigated DBA fairly distributes bandwidth based on each ONU's demand. Consequently, under low or balanced demand conditions, the granted transmission windows tend to be similar across all ONUs. It is important to highlight that the same qualitative behavior is observed when T-CONT 1 or T-CONT 2 are also present in the network. In such cases, the bandwidth shared among the T-CONT 3 ONUs corresponds to the total upstream capacity minus the fixed bandwidth already allocated to the T-CONT 1 and T-CONT 2. Beyond data rate estimation, these observations show that BWMap analysis enables implicit network profiling: allocation patterns reveal DBA behavior, and T-CONT types can be inferred from Alloc-ID values.

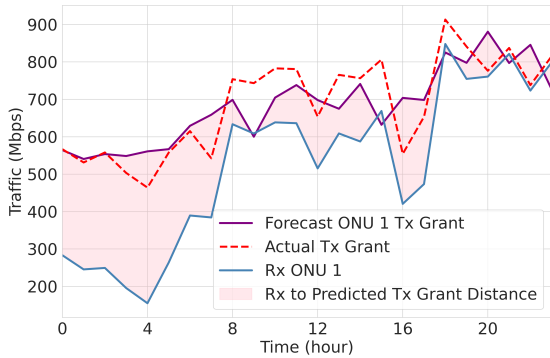


**Fig. 2:** Input traffic profiles for ONU 1 and ONU 2 during the measurement campaign.

With these insights into the network behavior, we proceeded to evaluate actual congestion estimation using data collected over a 5-day campaign on the testbed. Three ONUs were configured with T-CONT 3, each with a PIR of 1.24 Gbps and CIR of 1 Mbps. Raw samples were collected at each traffic injection point and subsequently processed.

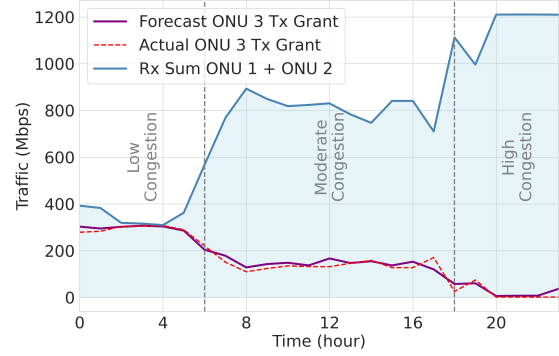
Using the BWMap recorded at every sampling instant, we extracted the sequence of transmission grants assigned to each ONU per allocation cycle. Let  $g_t$  denote the size of the transmission window (in bytes) assigned at cycle  $t$ . Since these grants evolve sequentially over time and may exhibit recurring daily patterns due to periodic traffic loads, we modeled  $\{g_t\}$  as a time series. A Seasonal Autoregressive Integrated Moving Average (SARIMA) [11] model was selected because it can capture short-term temporal dependence as well as seasonal behavior with a relatively simple and interpretable formulation. The upstream traffic injected by the first two ONUs is shown in Fig. 2. The third ONU was configured with a constant low bitrate of approximately 1 Mbps and was therefore omitted from the figure due to the large difference in scale compared to the other two ONUs.

The per-hour traffic patterns were inspired by the literature [12–14] but normalized to our system capacity. Four days traffic points were injected into the PON, and the corresponding BWMap was retrieved for each traffic point, as detailed previously. From the transmission grants (Tx Grant) assigned to the three ONUs over the four days, a SARIMA model was developed to capture the behavior of the granted windows. Although the dataset was relatively limited, it proved sufficient to model the underlying periodic behavior, with each 24-hour interval defined as one seasonal period.



**Fig. 3:** Comparison between predicted Tx Grant and received traffic (Rx) for ONU 1 over a representative day.

The predicted Tx Grant values enabled an analysis of the relationship with the received traffic (Rx). Fig. 3 shows this comparison for ONU 1. Three distinct regimes can be observed. Until approximately 6:00 AM, the traffic demand of all ONUs remained well below the network's maximum capacity (1.24 Gbps). Consequently, the DBA algorithm over-allocated bandwidth to the ONUs and the congestion estimation from the PON frames is less faithful to the network state. Between 6:00 AM and 6:00 PM, the Tx Grant approached the actual capacity limits more closely. Finally, after 6:00 PM, as the aggregate demand neared the network's capacity limit, the predicted Tx Grant and the actual



**Fig. 4:** Tx Grant assigned to ONU 3 and its relationship with network congestion level (represented by the aggregate traffic of the other ONUs).

granted bandwidth converged closely.

Additionally, it can be observed that the third ONU, despite transmitting very low traffic volumes, provided valuable insight into network behavior. As shown in Fig. 4, the Tx Grant assigned to ONU 3 exhibits an inverse relationship with the overall network congestion level. Three distinct flat levels are clearly visible on the graph, corresponding to the three primary load regimes of the network.

Under the tested conditions, ONU 3 effectively served as a monitor node for network load. Furthermore, the SARIMA-based estimation of the Tx Grant closely matched the actual granted values.

## Conclusions

This work demonstrates that the upstream BWMap embedded in GPON downstream frames constitutes a powerful signal that enables fully passive, vendor-agnostic monitoring of upstream bandwidth utilization. By decoding transmission grants, we have shown that network congestion and achievable upstream throughput can be accurately estimated and forecasted without any access to OLT management systems or user traffic data.

Experimental results from our commercial-grade testbed confirm that the Tx Grant serves as a reliable proxy of actual received traffic, particularly under medium-to-high load regimes, while SARIMA modeling successfully captures daily congestion patterns and identifies low-utilization windows. These capabilities enable practical applications such as SLA verification, DBA behavior profiling, and opportunistic service scheduling in live PON deployments.

The main limitation observed is the over-allocation behavior of the DBA under low-load conditions, which reduces estimation accuracy. This motivates future research on adaptive, load-aware models and the extension of the approach to multi-vendor and multi-service scenarios. Overall, BWMap-based passive monitoring opens a new, low-cost pathway toward transparent and intelligent management of PON infrastructures.

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